

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal (CPR)
| LECTURE 6

Clinical Embryology of the Heart

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Recommended Reading

- Moore, Persaud, Torchia. The developing human. Clinically oriented embryology 11th Ed.
- Printed Textbook: Chapter 13, Pgs. 263 - 291.
- Online Textbook: Chapter 3: Development of Cardiovascular system
- Pay special attention to the **Green/ blue** boxes for the clinical correlations for this topic.

Images taken from:

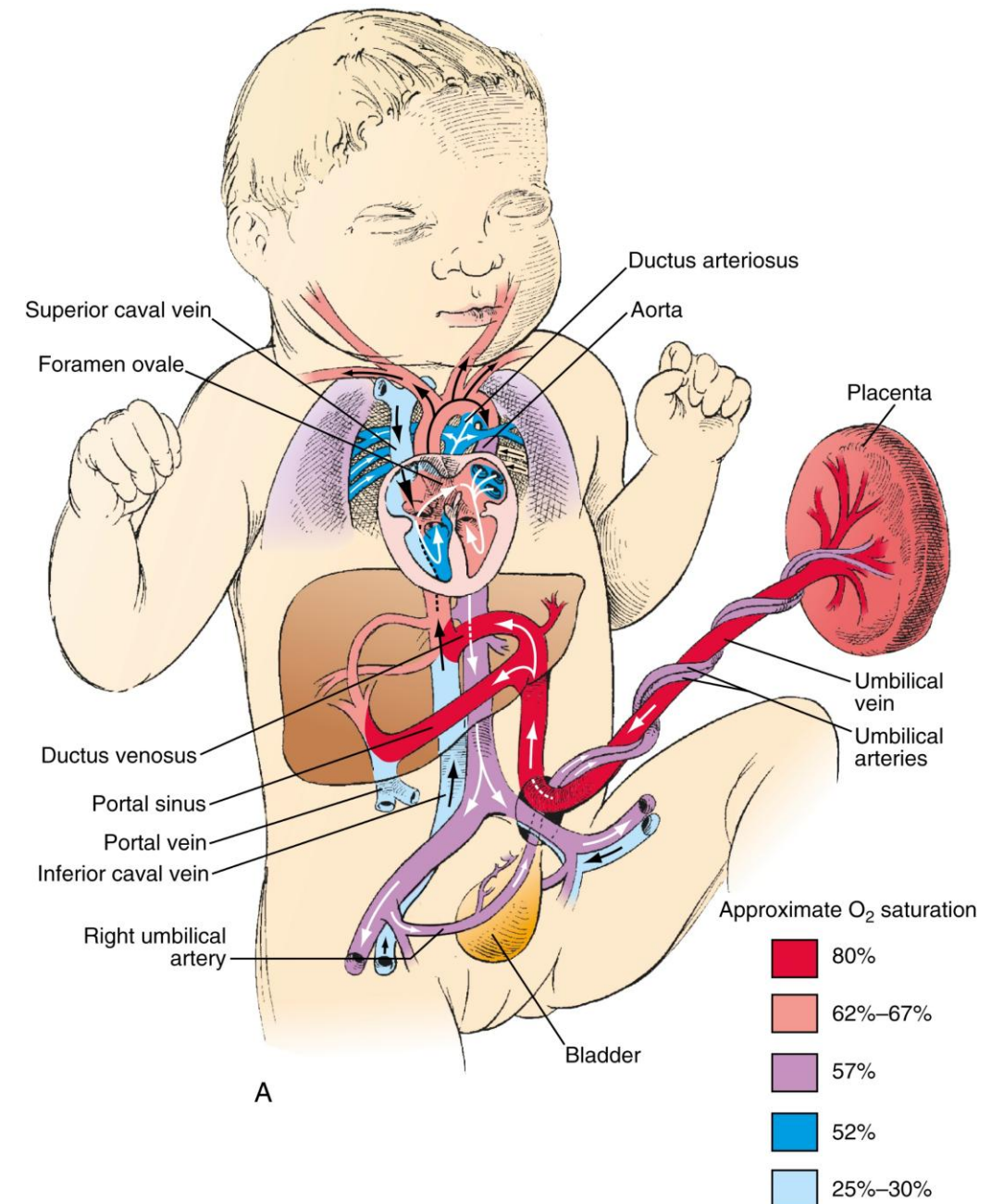
- The Developing Human Clinically Oriented Embryology, 11th Edition. Keith L. Moore, T.V.N. Persaud, Mark G. Torchia. *Elsevier*
- Langman's Medical Embryology, 14th Edition. T.W. Sadler. *Wolters Kluwer*
- Peter H. Abrahams, Jonathan D. Spratt – McMinn's & Abrahams' Clinical Atlas of Human Anatomy, 7th Edition (2013, Mosby).
- Atlas of Human Anatomy, 6th Edition – Frank H. Netter, M.D.

Objectives

SOM.MK.II.BPM1.3.CPR.2.ANAT.1416	Describe fetal circulation and discuss the changes that occur at birth.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1417	List the most common heart defects in Down's syndrome.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1418	List left-to-right shunts in congenital heart defects and explain why they are generally acyanotic.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1409	Describe five types of atrial septal defects.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1412	Describe the two types of ventricular septal defect.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1419	List right-to-left shunts in congenital heart defects and explain why they are generally cyanotic.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1420	Describe the developmental mechanisms underlying ectopia cordis, persistent truncus arteriosus, and transposition of the great arteries, tetralogy of Fallot, total anomalous pulmonary venous return and patent ductus
SOM.MK.II.BPM1.3.CPR.2.ANAT.1349	Describe the embryologic origin of coarctation of the aorta.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1350	Explain how coarctation of the aorta results in a difference between femoral and radial pulses and notching of the ribs.

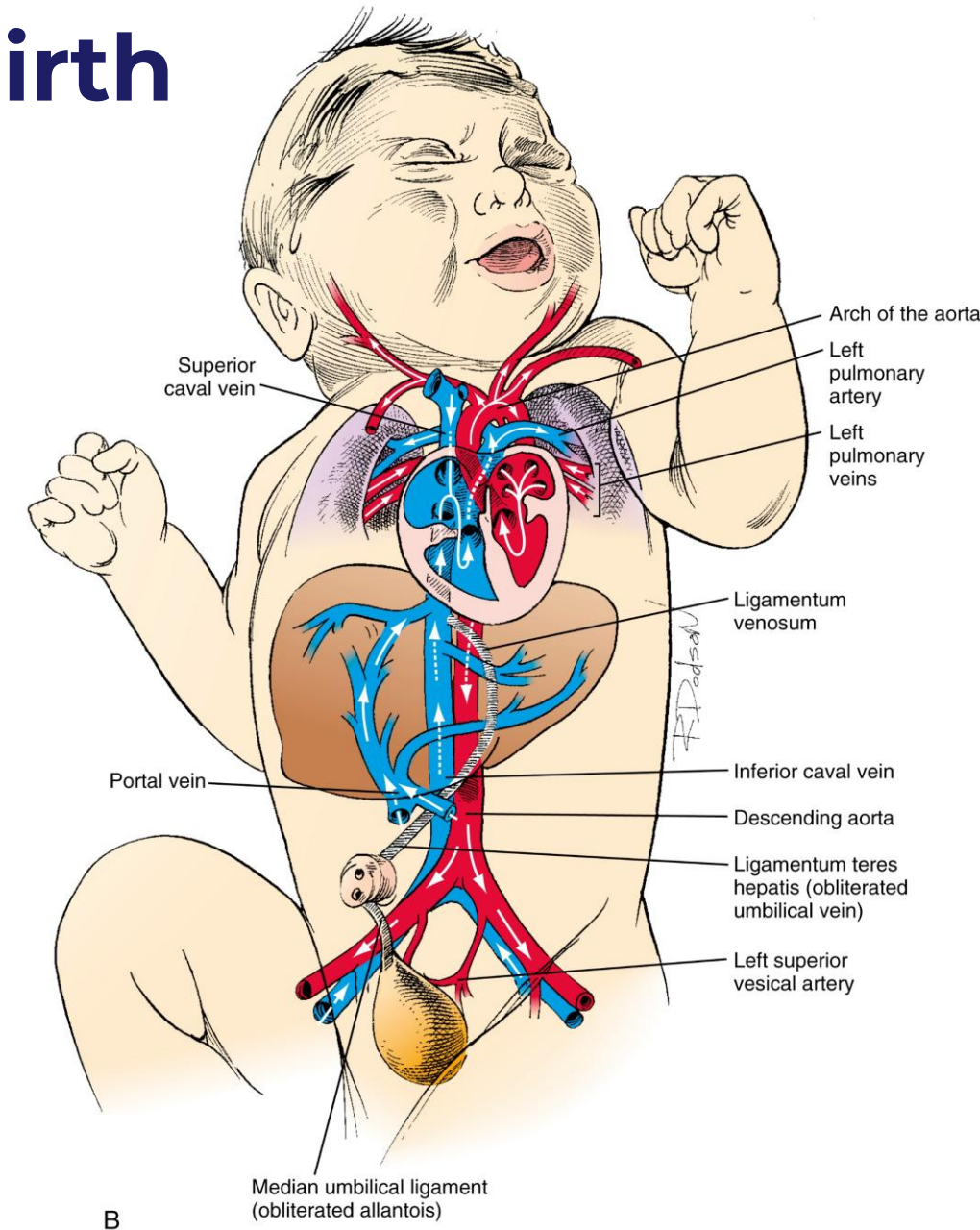
Fetal circulation

- Blood enters the **right atrium** from:
 - **Inferior vena cava**; **deoxygenated blood** from the fetus and **oxygenated blood** from the placenta via umbilical vein
 - **Superior vena cava**; **deoxygenated blood** from fetus
- Right atrium sends blood to:
 - Right ventricle
 - **Most of the blood gets shunted to the left atrium through foramen ovale**
- Right ventricle sends blood to the lungs;
 - **Ductus arteriosus** (connecting the pulmonary trunk and aorta) **shunts** most of the blood **into the aorta**
- Left atrium sends blood into the aorta
- Aorta goes to the fetal body
 - Umbilical arteries go to the placenta



Changes in circulation at birth

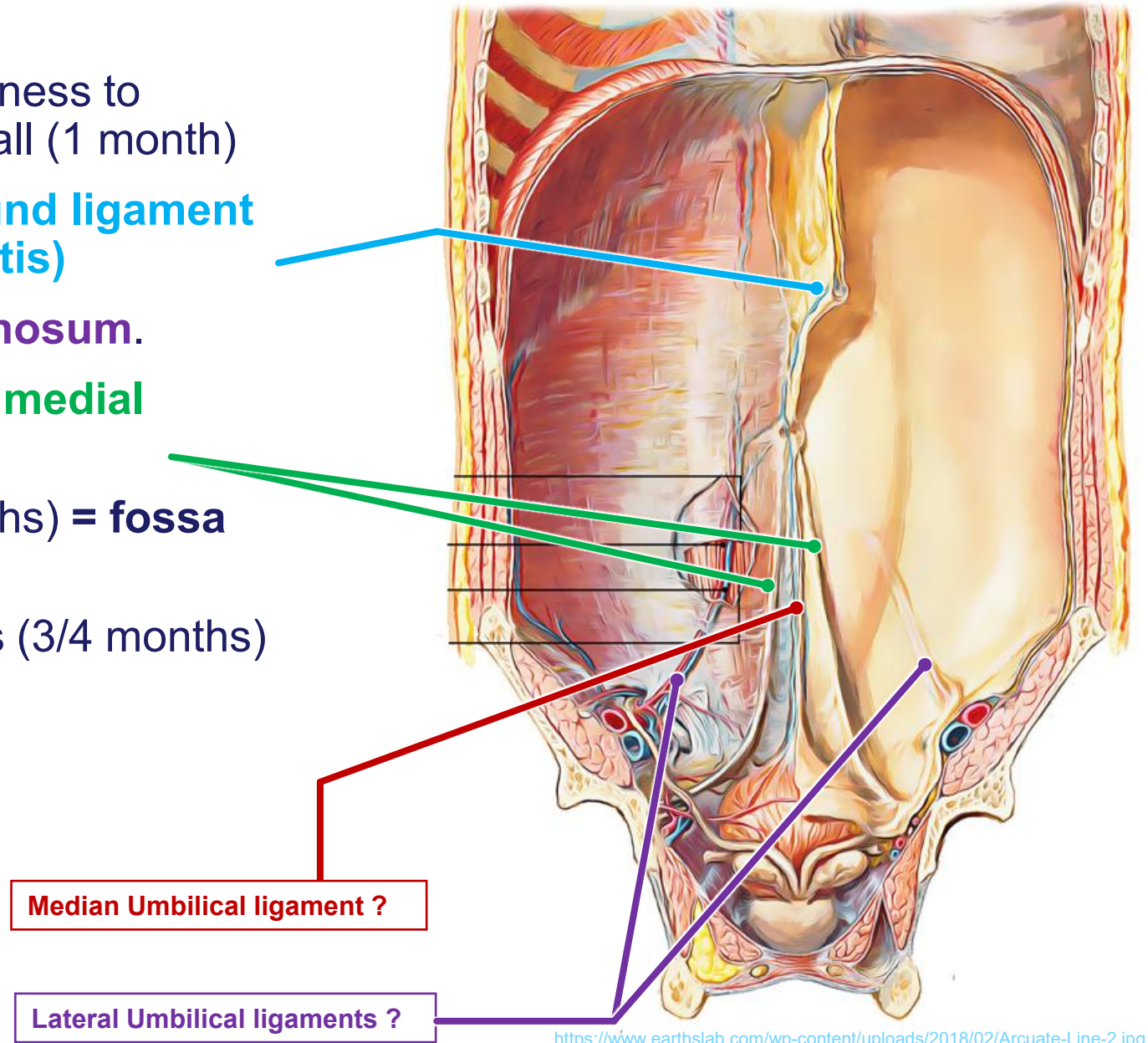
- Aeration of lungs at birth causes a **decrease in pulmonary vascular resistance**
 - Allows blood to flow fully into the lungs
- Loss of flow from umbilical cord causes right atrial pressure to **drop** lower than left atrial pressure
 - **Functional closure of foramen ovale at birth**
- Pulmonary vascular resistance **drops** below systemic vascular resistance causing reversal of ductus arteriosus
 - **Functional closure ductus arteriosus at birth** mediated by high oxygen tension **decrease in prostaglandin E** leading to vasoconstriction



B

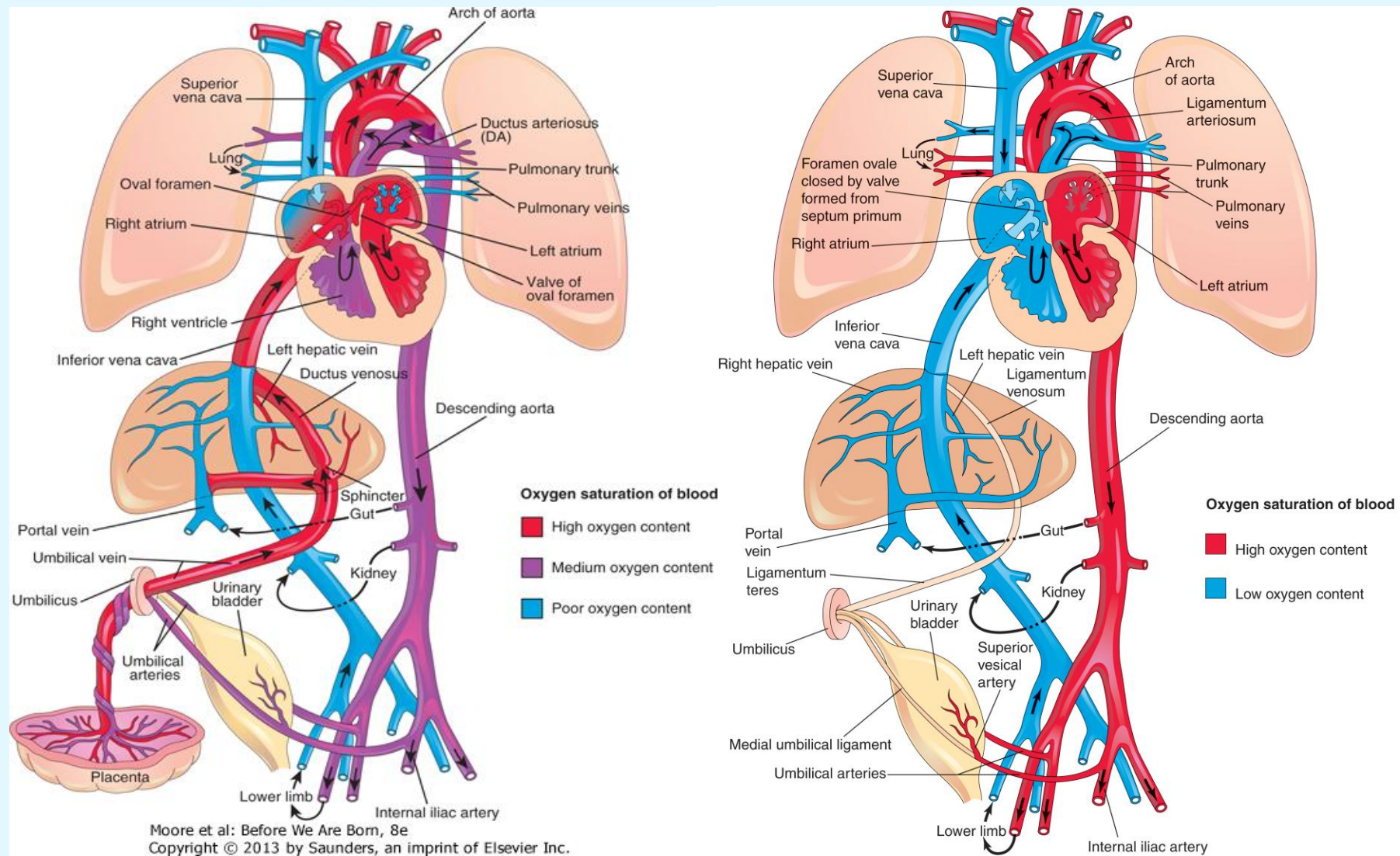
Changes after birth

- Left ventricular wall increases in thickness to surpass that of the right ventricular wall (1 month)
- Intraabdominal **umbilical vein** → **round ligament of the liver (ligamentum teres hepatis)**
- **Ductus venosus** → **ligamentum venosum.**
- Intraabdominal **umbilical arteries** → **medial umbilical ligaments.**
- Full closure of foramen ovale (3 months) = **fossa ovalis**
- Anatomic closure of ductus arteriosus (3/4 months) = **ligamentum arteriosum.**



Fetal circulation vs Neonatal circulation

- https://www.clinicalkey.com/student/content/video/23-s2.0-mm_9780323696043_0041
- <http://drquike.blogspot.com/2012/12/fetal-circulation.html>



Congenital Heart Defects (CHD)

- **Common** – about 6 – 8 per 1000 births
- Well-tolerated in fetal life
- After birth, can be **asymptomatic** (discovered incidentally), **symptomatic** (e.g., **Cyanosis**), or **Incompatible with life**.
- **Etiology** (~80% is unknown)
 - **Infections** e.g., rubella - early pregnancy
 - **Chromosomal** abnormalities:
 - Down's syndrome
 - Turner syndrome
 - DiGeorge syndrome or conotruncal anomaly face syndrome (chromosome 22 deletions)
 - **Environmental** factors:
 - Chemical agents
 - Alcohol
 - Drugs (illicit and prescription)
 - Maternal diabetes

CHD - Left To Right Shunt (Acyanotic)

- Defect leading to “Back-leak” of blood from systemic (high pressure) to pulmonary circulation (low pressure)
 - Left atrium - right atrium**
 - Atrial septal defect (ASD)
 - Patent foramen ovale (PFO)
 - Left ventricle - right ventricle**
 - Ventricular septal defect (VSD)
 - Aorta - pulmonary trunk**
 - Patent ductus arteriosus (PDA)
 - Both atria and ventricle**
 - Atrioventricular septal defects (AVSD)
- ACYANOTIC (without cyanosis) → No flow of deoxygenated blood through the systemic circulation
- Pulmonary flow will be larger → **Pulmonary hypertension**
- Heart failure** (No enough blood to aorta)

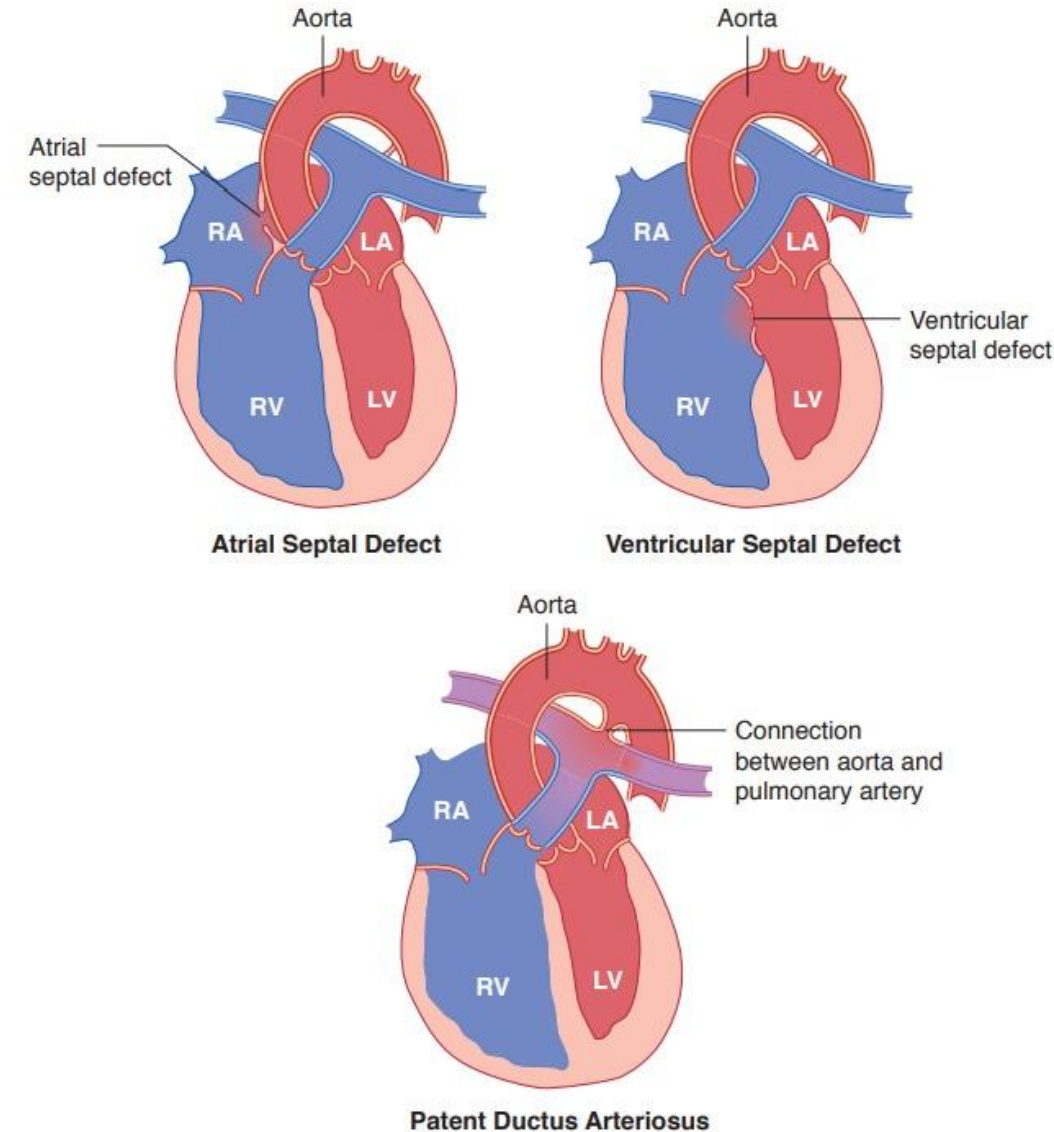
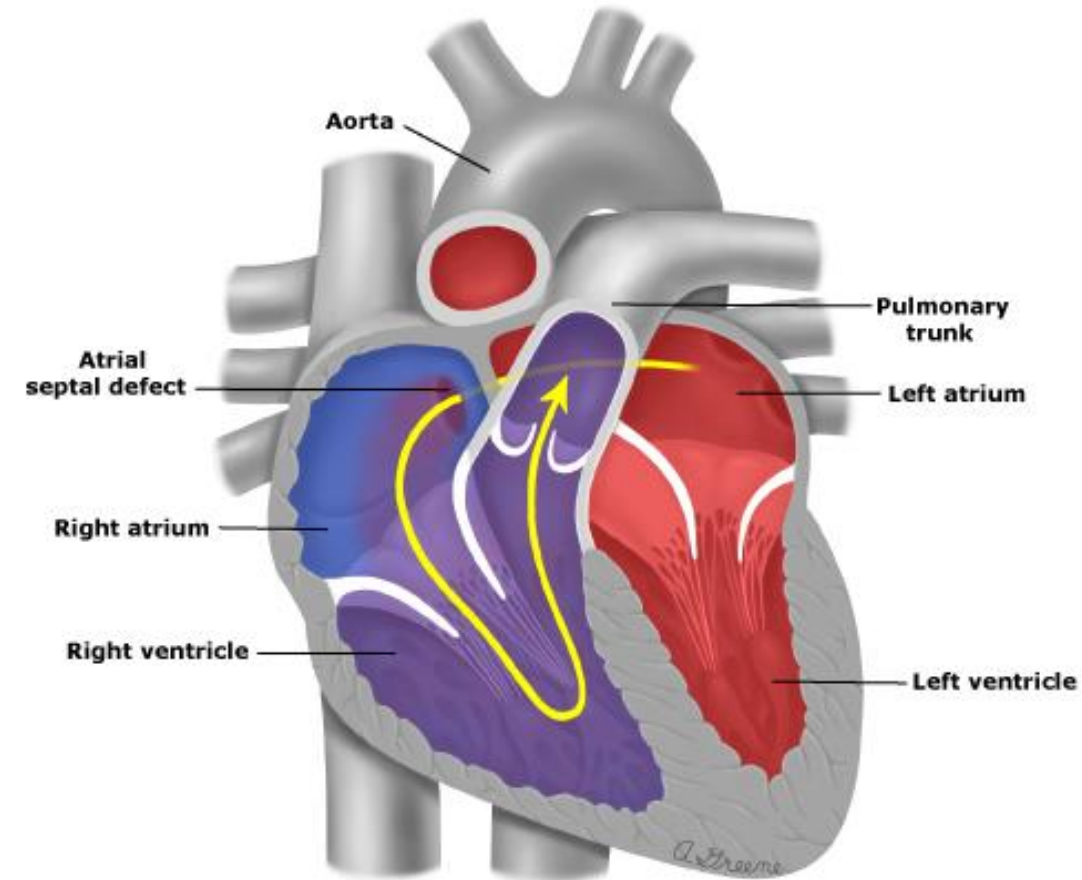
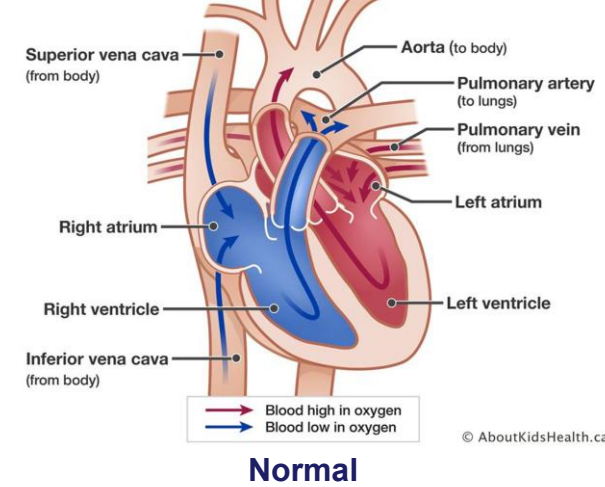


Figure 13-4. Common Forms of Acyanotic Congenital Heart Disease

Left To Right Shunt - Atrial Septal Defects

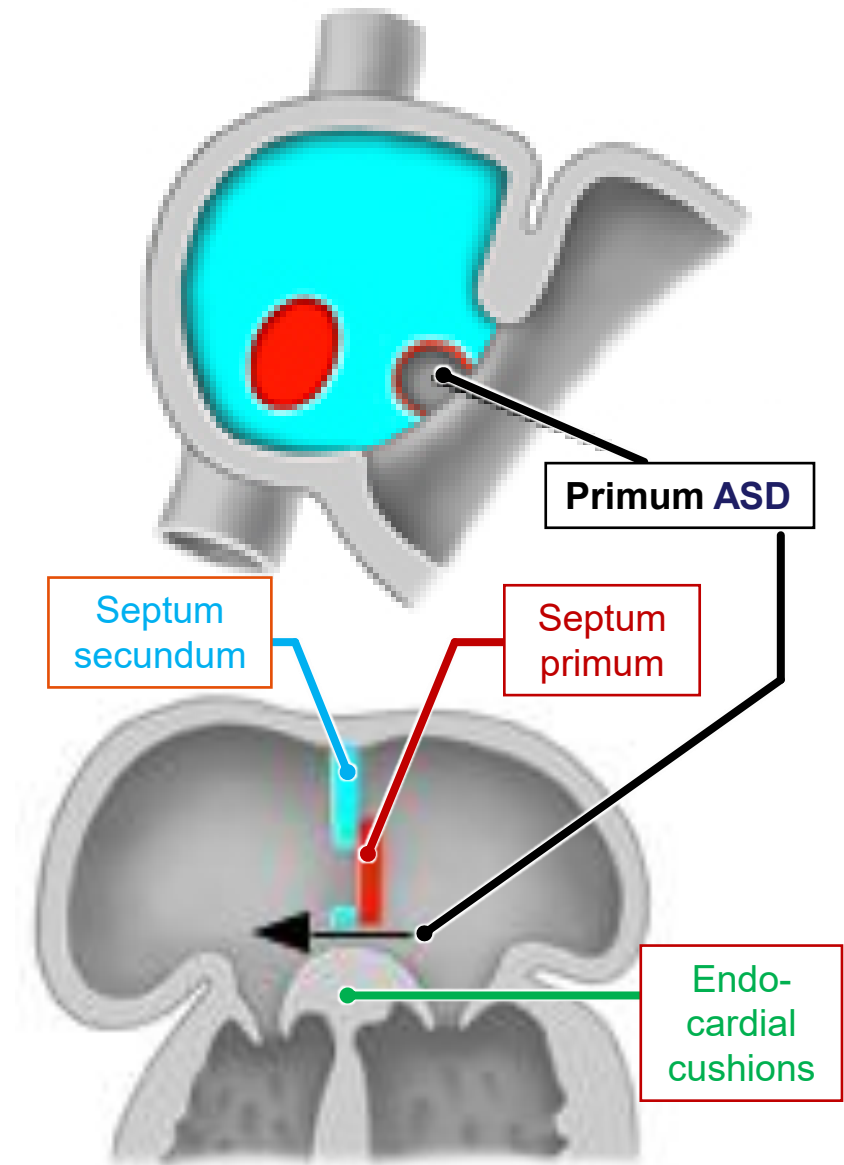
- 10 – 15 % of CHD
- 1 to 2 per 1000 live births.
- ASDs are based on embryogenesis defects into:
 - Primum ASD
 - Secundum ASD
 - Sinus venosus ASD
 - Common atrium

Clinical Significance:
Left to right shunt after birth
(acyanotic heart defect)



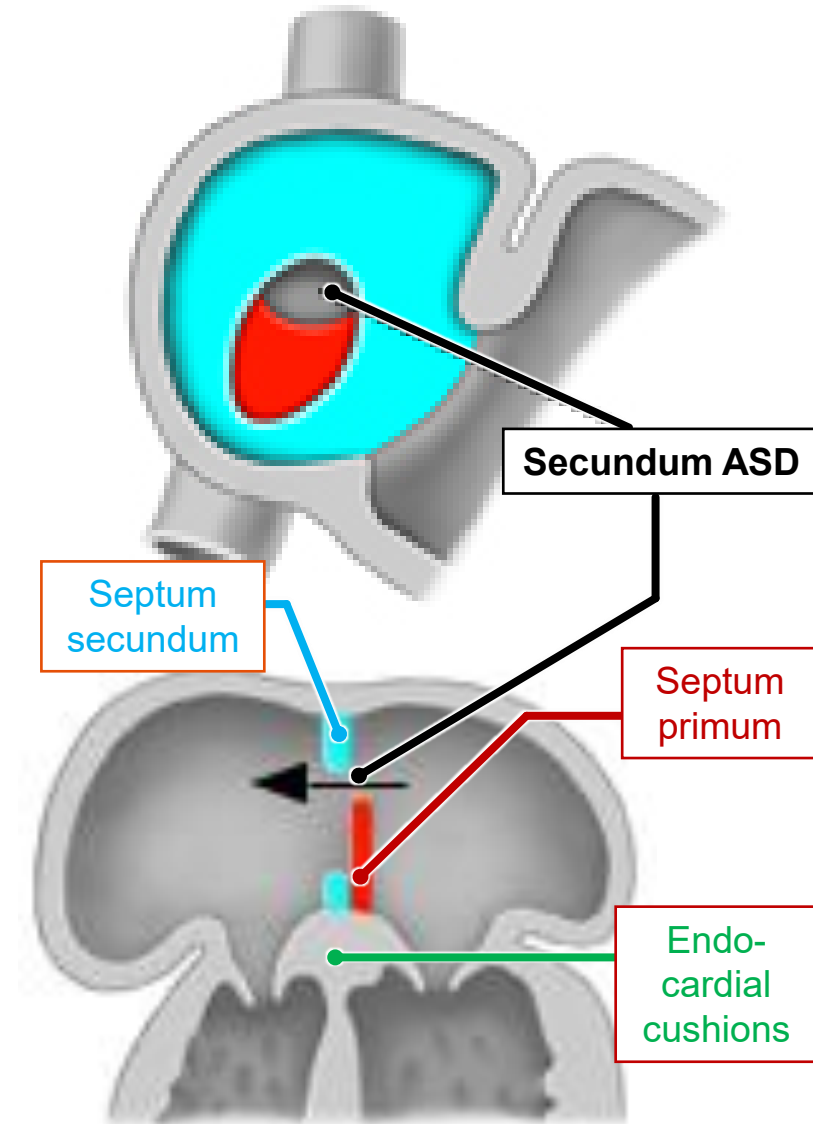
Atrial Septal Defects - Primum ASD

- Syn. *Ostium/foramen primum defect*
- Relatively rare.
- **Septum primum does NOT** fuse with endocardial cushions, leaving a defect at the base of the interatrial septum
- Associated with absence or inadequate formation of endocardial cushions
- Typically associated **with AV canal defects**:
 - AV valve defect (cleft in the anterior cusp of the mitral valve)
 - Ventricular septal defect



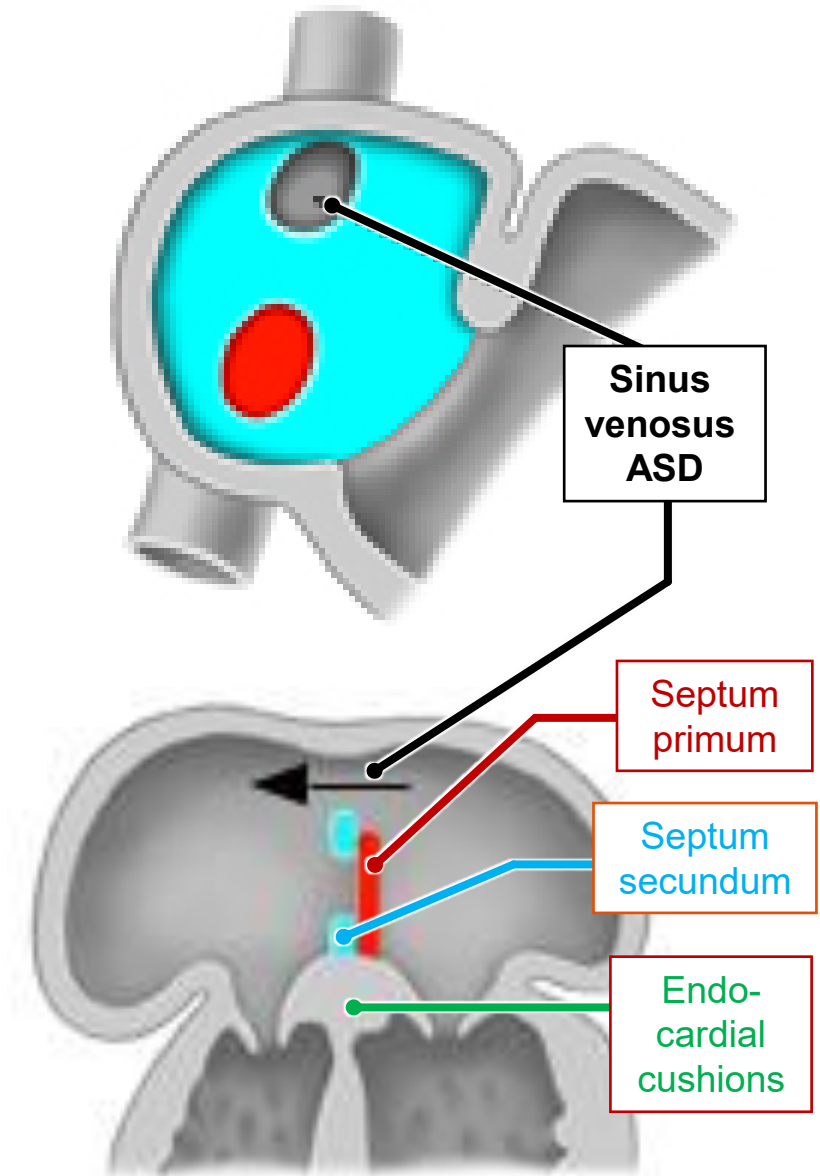
Atrial Septal Defects - Secundum ASD

- Syn. *Ostium/foramen secundum defect*.
- Commonest type (70%) but least severe
- Female to male ratio: 3:1
- Caused by **arrested growth of the septum secundum** or **excessive apoptosis of the septum primum**, leading to a large open foramen secundum
- Associated with:
 - Holt-Oram syndrome: (Autosomal Dominant)
 - Upper limb defects + secundum ASD
 - Mitral valve prolapse
- Mostly **asymptomatic** → commonly undiagnosed in infancy and childhood, and spontaneously resolve



Atrial Septal Defects - Sinus venosus ASD

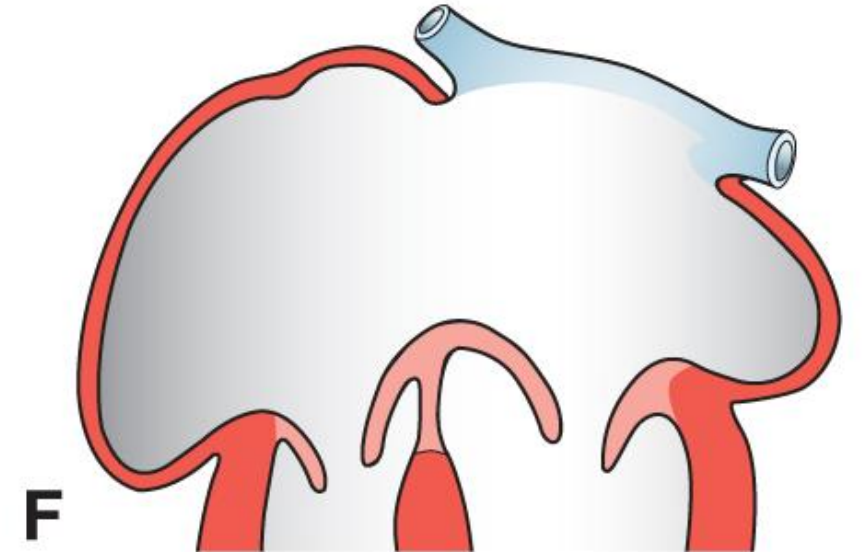
- A rare type of ASD located in the **superior part of the interatrial septum**
- Defect:
 - **Incomplete absorption of the right horn** of sinus venosus into the right atrium
 - **Abnormal development of the septum secundum**, as it normally arises from the superior wall of the right atrium.
 - or a combination of these factors
- Located **near the orifice of the superior vena cava** → partial anomalous pulmonary venous connection



Atrial Septal Defects - Common atrium/Cor Triloculare biventriculare

- A rare type of ASD where the **interatrial septum is completely absent**
- Defect
 - Failure of development of the septum primum, septum secundum and sinus venosus
 - Leading to a combination of primum, secundum and sinus venosus defects
 - It is the most severe type of ASD.

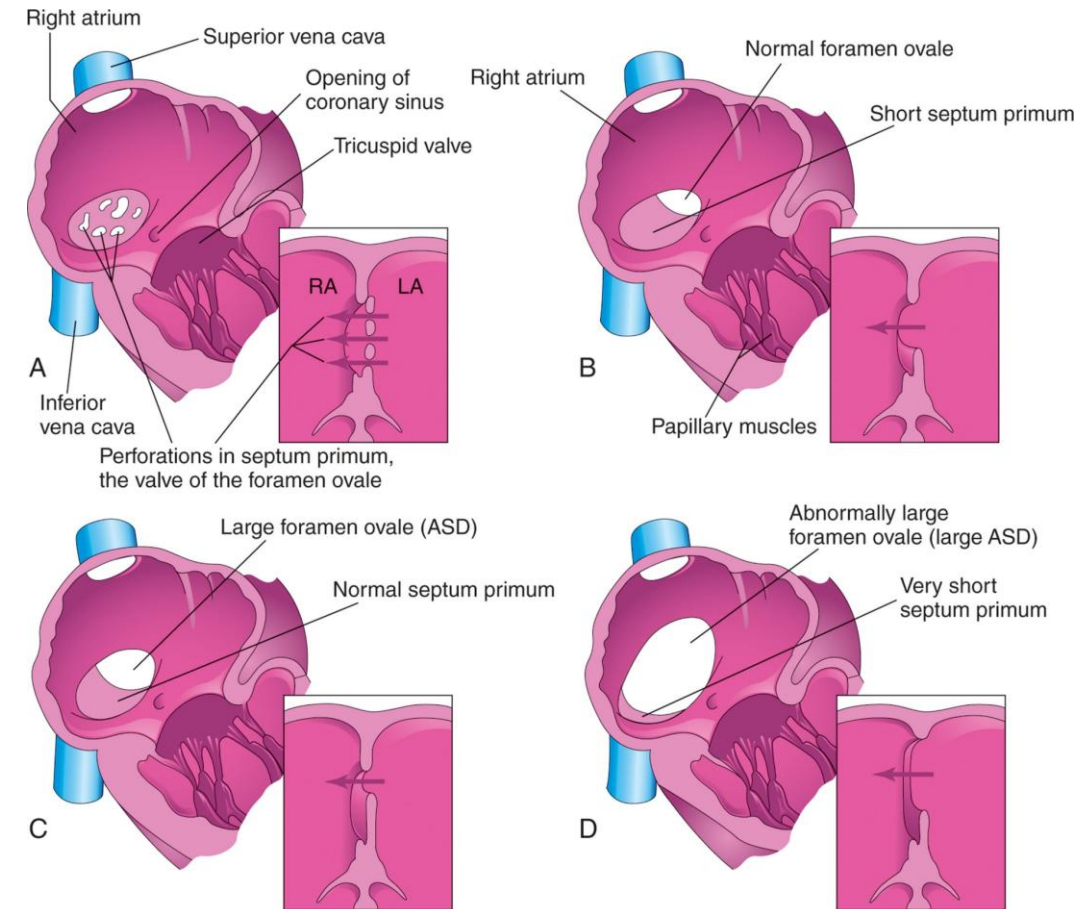
Often associated with defects in other locations, e.g., malformations in the AV valves, VSD.



Absence of septum primum and septum secundum

Left To Right Shunt - Patent Foramen Ovale

- Diameter = 1 to 10 mm
- **25 – 30 % of adult population**
- Occurs when there is an **incomplete adhesion of the flap-like flexible valve** (septum primum) with the rigid septum secundum after birth.
- Interatrial shunting generally does not occur as long as left atrial pressure exceeds right atrial pressure
- Most individuals **remain asymptomatic**.
- Rarely associated with
 - Cryptogenic stroke
 - Migraine and vascular headache
 - Decompression sickness and air embolism
- An incidentally detected PFO generally requires no follow-up or treatment.



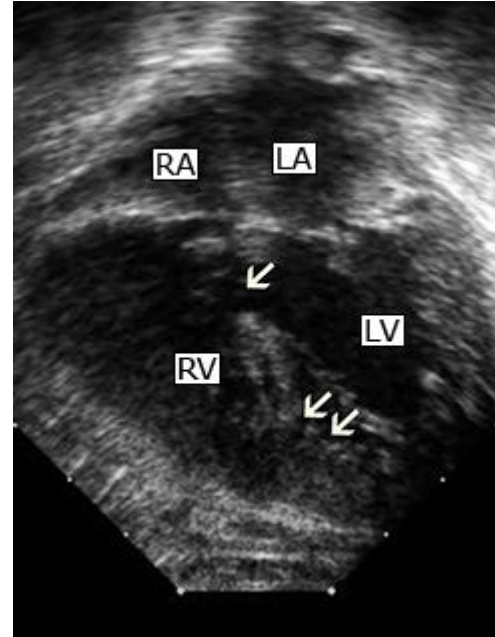
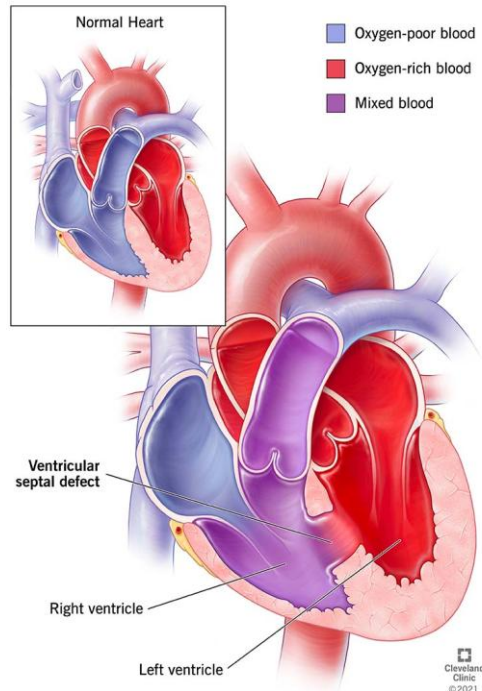
Left To Right Shunt - Ventricular Septal Defect (VSD)

- Most common (25% of all CHD)
- Males > females
- Two types: Muscular & Membranous
- Clinical Significance:
 - Both will lead to left to right shunt after birth (acyanotic heart defect)
 - When large → dyspnea and heart failure in early infancy.

Left To Right Shunt - Ventricular Septal Defect (VSD)

Muscular VSD

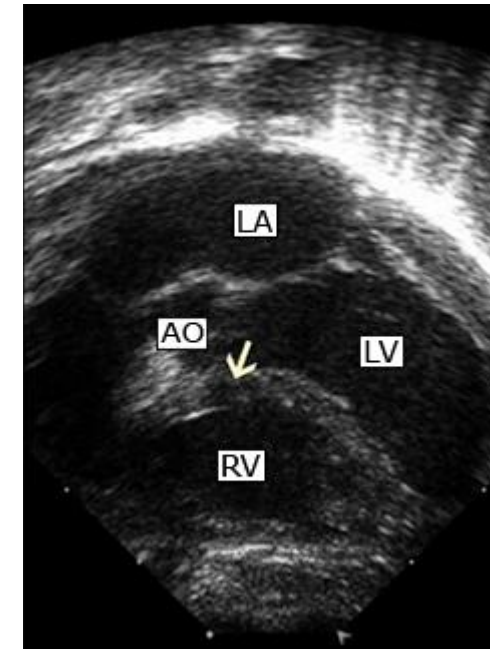
- Anywhere on the muscular part of the IV septum
 - A single location or, multiple small defects along the septum (“Swiss Cheese” VSD)
- Complete absence of the IV septum leads to a single/common ventricle (3-chambered heart/cor triloculare biatriatum) that can result in serious congestive heart failure and death.



Muscular VSD

Membranous VSD

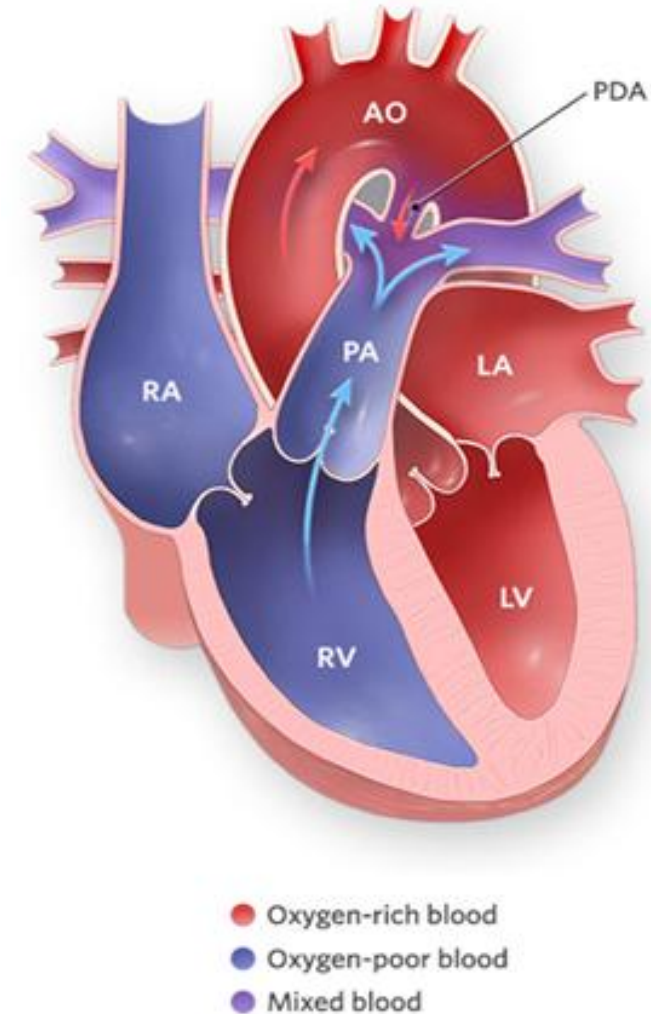
- Most common type
- Absent membranous interventricular septum caused by failure of endocardial cushions to develop and fuse with the muscular and aorticopulmonary septum



Membranous VSD

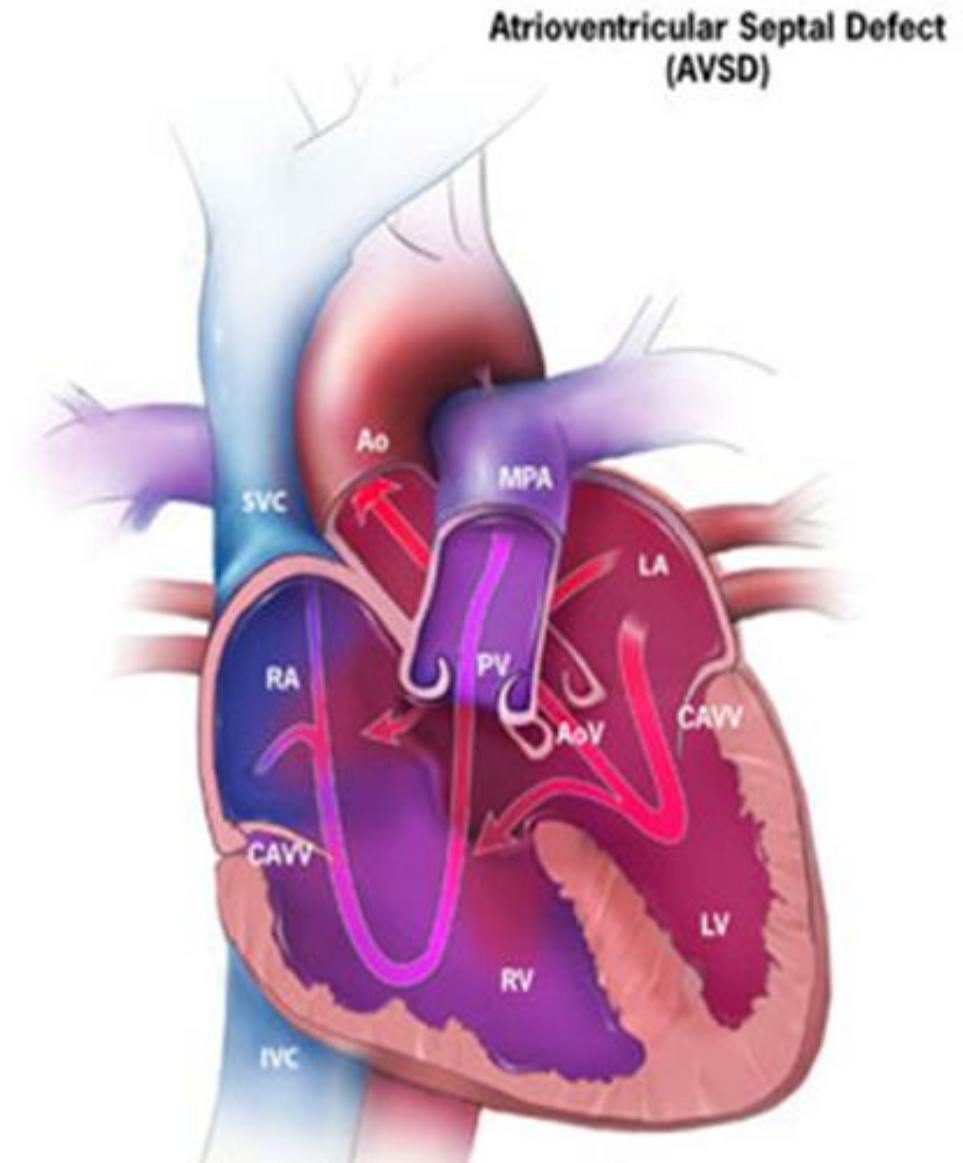
Left To Right Shunt - Patent Ductus Arteriosus (PDA)

- Females : males (2:1)
- Associated with maternal **rubella infection** in early pregnancy & **Down syndrome**
- Failure of the ductus arteriosus to involute due to **failure of contraction of the muscular wall** of the ductus
- Common in: Premature infants, infants at high altitudes, hypoxia and low birth weight (<1750g)
- Clinical Significance:
 - **Difficulty breathing, feeding problems, poor growth, lung infections**
- Can be treated by anti-prostaglandins (NSAID - indomethacin) - Surgery (cardiac catheterization)



Left To Right Shunt - Atrioventricular septal defect (AVSD)

- Severe **endocardial cushion defect** where there is a large defect in the **atrioventricular wall**.
 - Leads to
 - Primum ASD
 - Membranous VSD
 - AV valve defects
 - In the most severe cases = Common AV valve
- Most common heart defect seen in **Down syndrome**



CHD - Right to Left Shunt (Cyanotic)

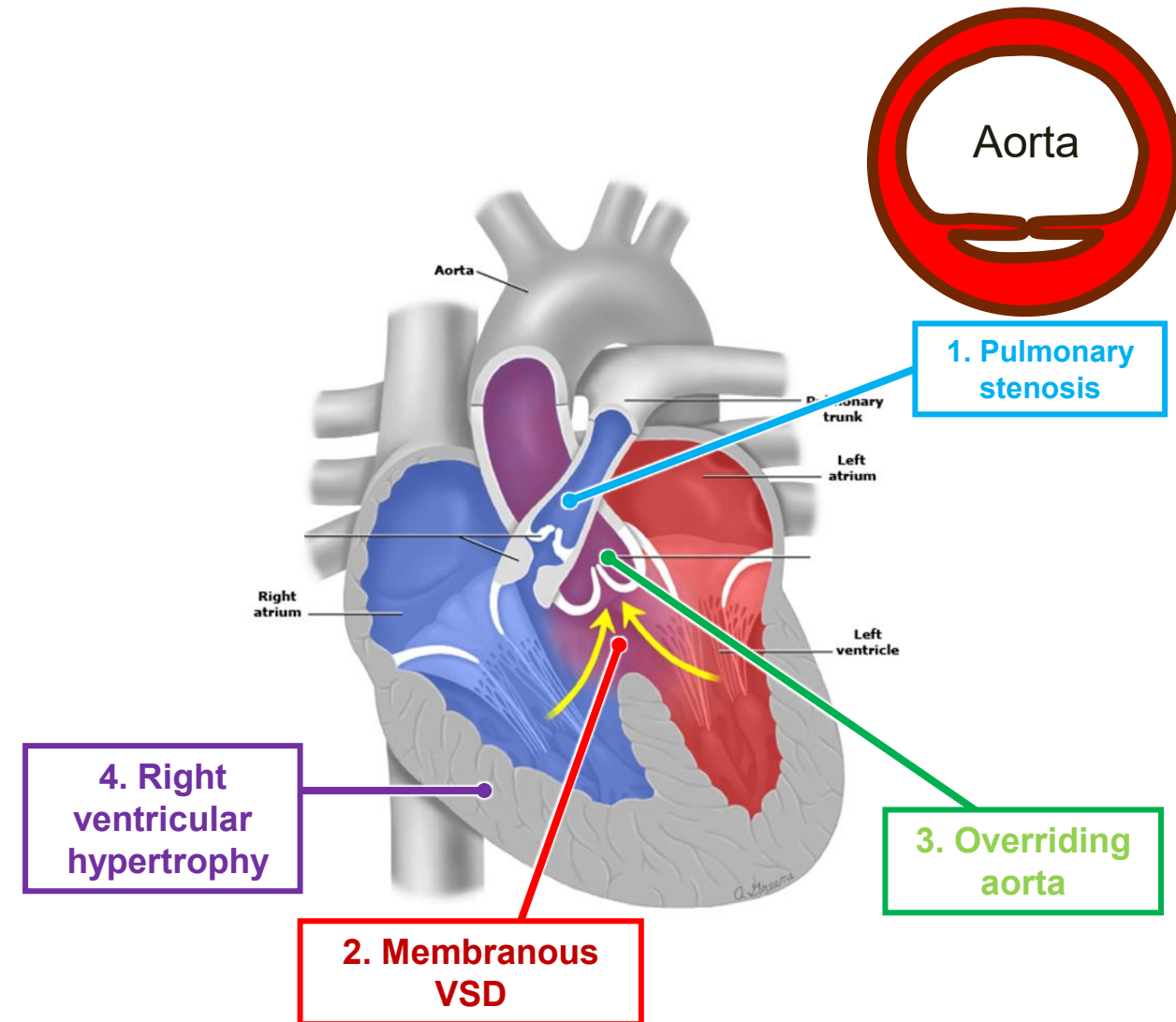
- Right to left shunts allow **deoxygenated systemic venous blood to bypass the lungs** and return to the body.
- Venous blood from the right heart (pulmonary circuit) **mixes** with the arterial blood in the left side of the heart.
- This decrease in oxygenated blood in the systemic circulation results in cyanosis.
- **CYANOSIS:**
 - Bluish discoloration of the skin and mucous membranes, due to excessive concentration of deoxyhemoglobin in the blood caused by deoxygenation.
- Usually requires surgical treatment
 - **Tetralogy of Fallot**
 - **Truncus arteriosus**
 - **Transposition of the great vessels**
 - **Tricuspid atresia**
 - **Total anomalous pulmonary venous return**

Detecting cyanosis in darker skinned babies

fingers, toes, tongue, under the tongue, lips, around the mouth
Also check areas that are normally pink

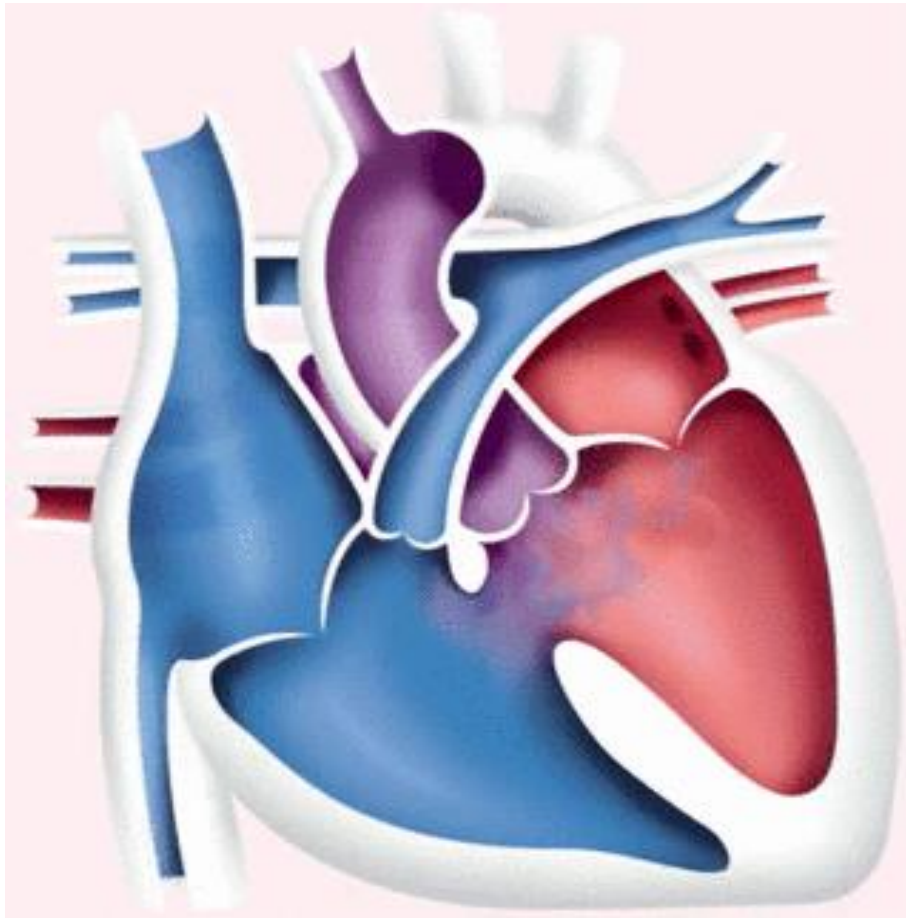
Right to Left Shunt (Cyanotic) - Tetralogy of Fallot

- **Pulmonary stenosis:** unequal division of the truncus arteriosus (a small (stenotic) pulmonary trunk)
 - anterior superior displacement of the bulbar (infundibular) septum
- **Membranous VSD** - failed fusion of the membranous septum components
 - malalignment of the truncal ridges, bulbar ridges and endocardial cushions
- **Overriding aorta** – combination of the pulmonary stenosis and membranous VSD causes the aorta to receive blood from both ventricles
- **Right ventricular hypertrophy** – due to increased workload resulting from pulmonary stenosis



Tetralogy of Fallot

- Commonly seen in DiGeorge syndrome, Down's syndrome, and fetal alcohol syndrome
- Has been linked to **abnormal migration of neural crest cells**



Knee Chest Position for a Tet Spell

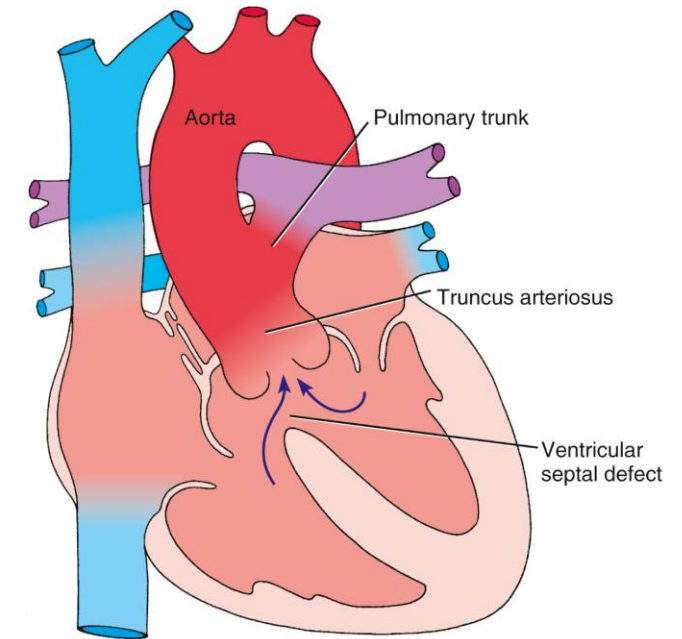


**“TET SPELLS” =
cyanosis with
vigorous crying,
feeding, or agitation**

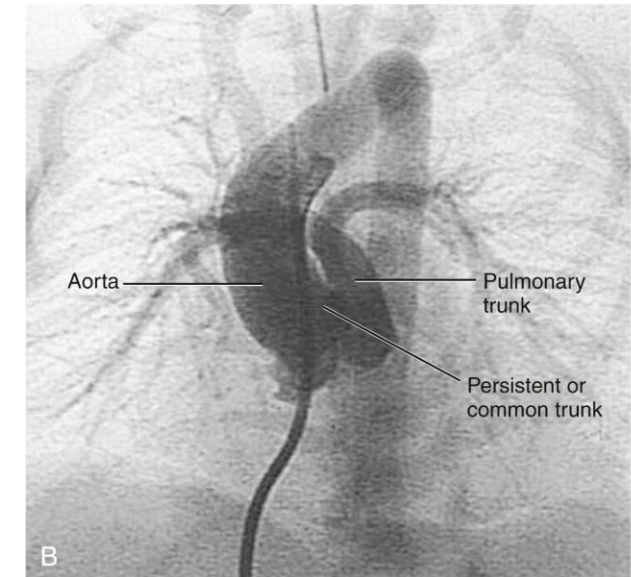
Knee chest position (squatting) improves the cyanosis
It **increases systemic vascular resistance**, promoting left-to-right shunting.

Persistent Truncus Arteriosus

- **Aortico-pulmonary septum does not form** = truncus arteriosus persists
 - Failure of neural crest cell migration
- **A VSD is always present** and the truncus arteriosus straddles the VSD
- Common defect in baby with microdeletion chromosome 22 (22q11) or DiGeorge syndrome
- Initially **MILD CYANOSIS**



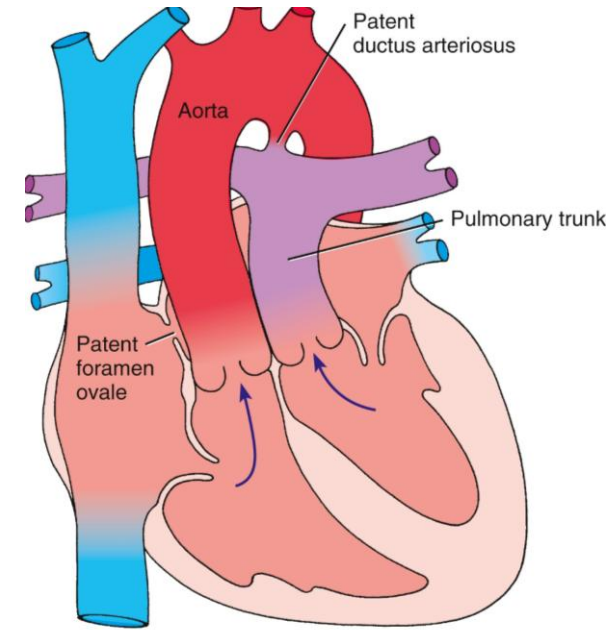
A Persistent truncus arteriosus



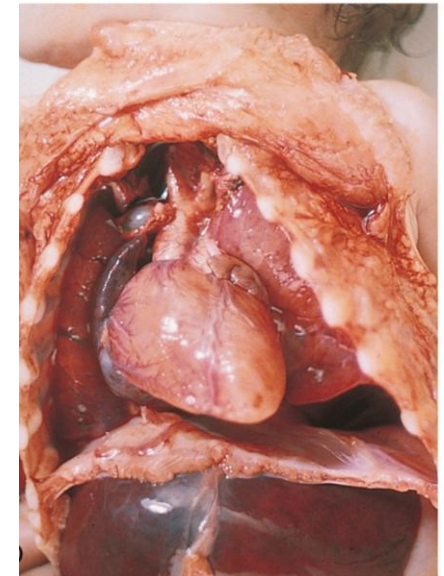
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Transposition of the Great Vessels

- Aorta and pulmonary trunk are transposed
 - Aorta connects with the R. Ventr and Pulmonary trunk with the L. Ventr
 - Faulty migration of **neural crest cells** leads to separation without spiraling
- This results in **SEVERE CYANOSIS** immediately at birth.
- Immediately lethal unless combined with another defect – **ASD, VSD or PDA.**
- Initial treatment by prostaglandin E1 (alprostadil) why?
- Surgical repair – The arterial switch operation (ASO)
- Frequently associated with maternal diabetes

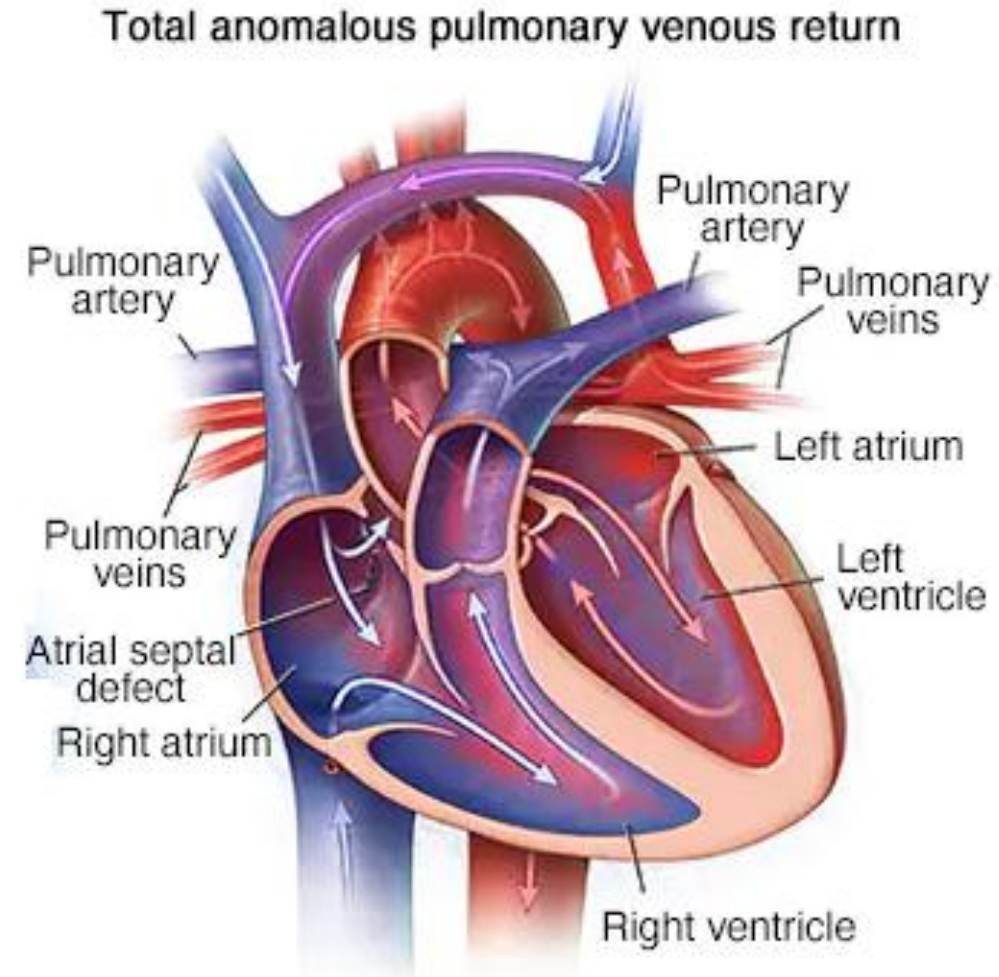


C Transposition of the vessels

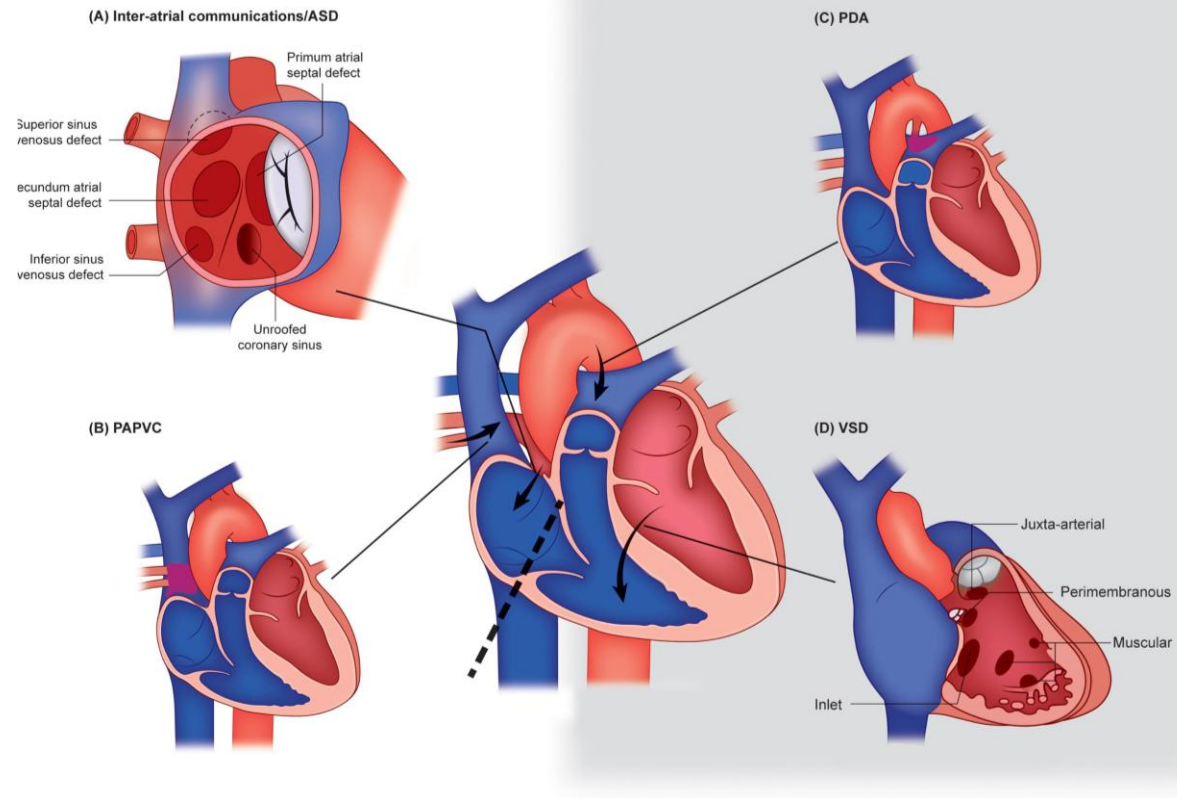


Total Anomalous Pulmonary Venous Return

- Abnormal drainage of the pulmonary veins into the systemic venous circulation. This results in **SEVERE CYANOSIS** immediately at birth.
- An **ASD, VSD or PDA** must exist to be compatible with life.
- Common sites:
 - **Superior vena cava +++++**
 - **Brachiocephalic vein +++++**
 - **Coronary sinus +++**
 - **Inferior vena cava +**
 - **Portal vein/hepatic vein +**

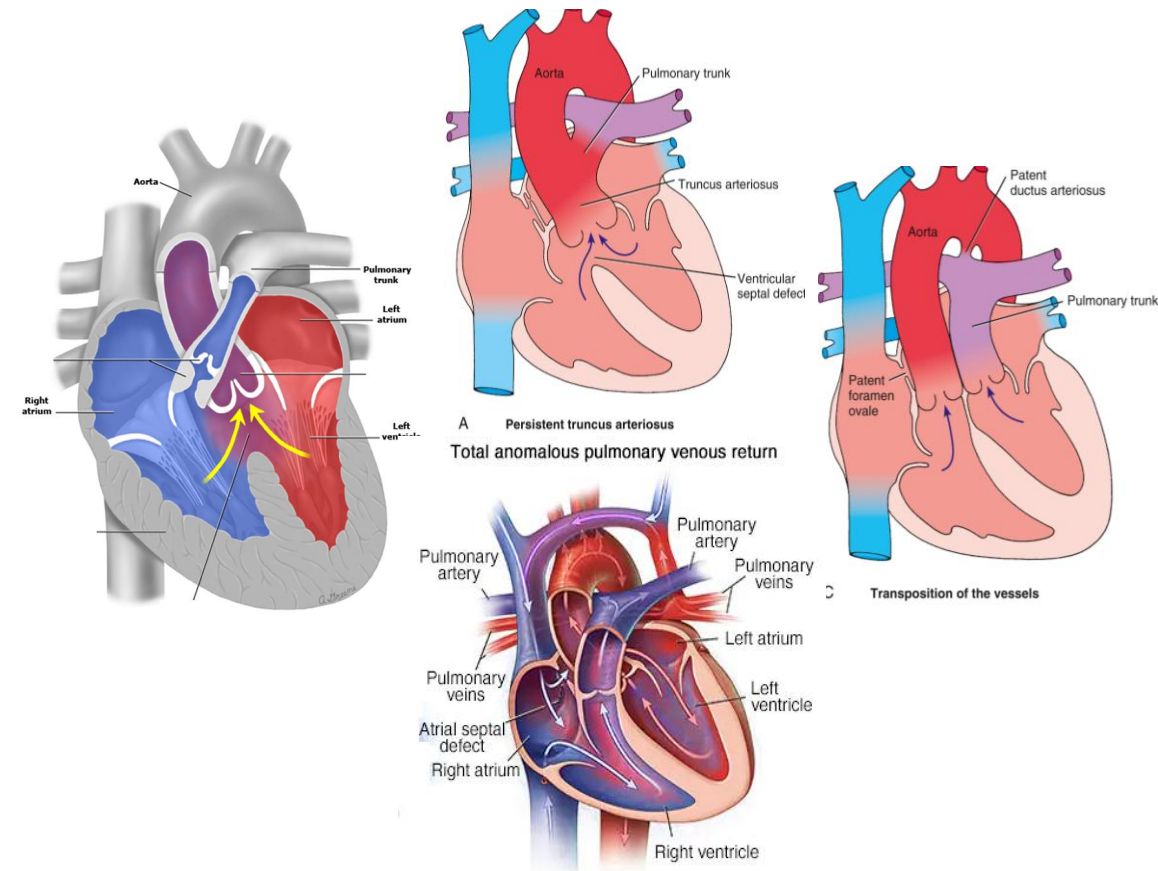


Cardiac shunts - Summary



Left-to-right shunts (Acyanotic)

1. Ventricular septal defects (VSD)
2. Atrial septal defects (ASD)
3. Patent ductus arteriosus (PDA)
4. Atrioventricular septal defects (AVSD)



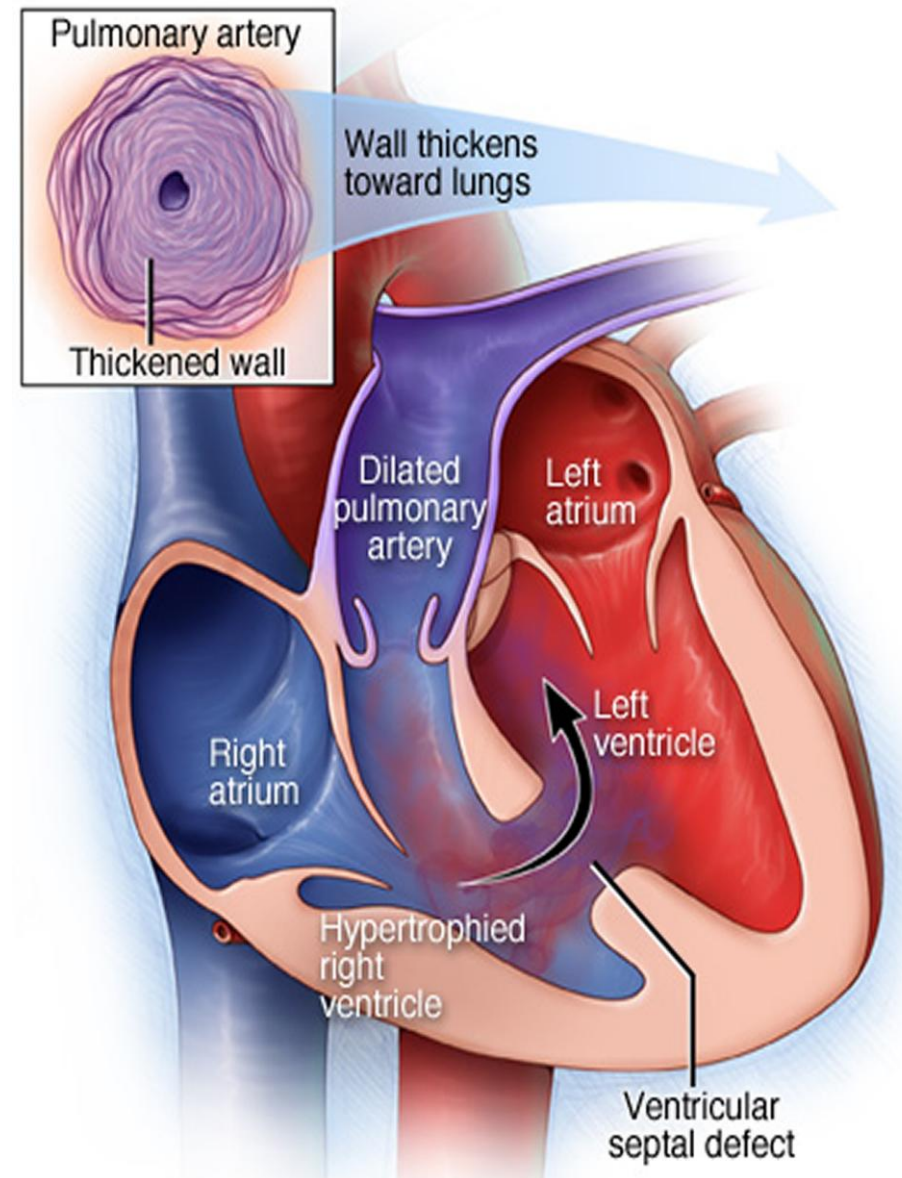
Right-to-left shunts (Cyanotic)

1. Tetralogy of Fallot
2. Truncus arteriosus
3. Transposition of the great vessels
4. Tricuspid atresia
5. Total anomalous pulmonary venous return

What happens when the left to right shunt is not repaired?

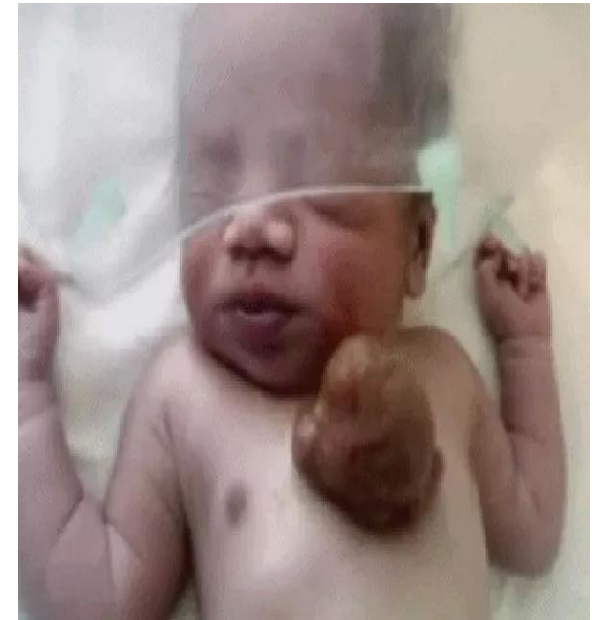
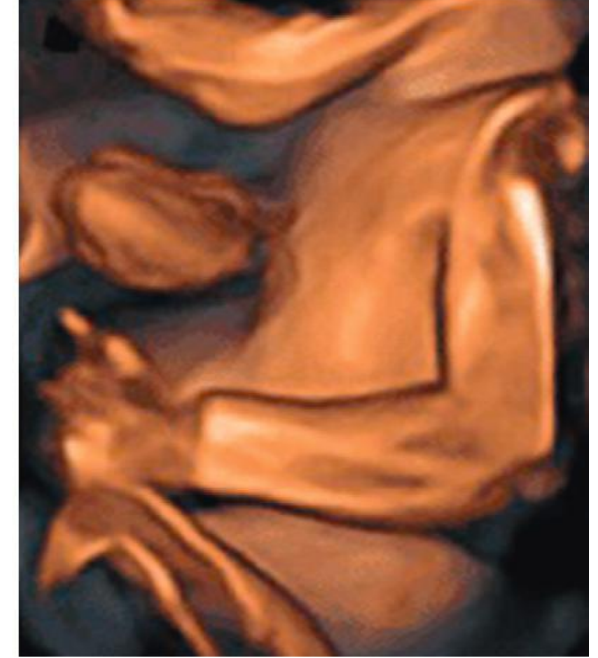
- Reversal of the shunt
- Large VSD, ASD, PDA and AVSD pushes excess blood into pulmonary circuit
 - Causes progressive damage to the pulmonary vasculature = **pulmonary hypertension**
- If untreated, the workload of right ventricle is increased = **right ventricular hypertrophy**
- Pressure in pulmonary circuit may ultimately **exceed** systemic pressure → reversal shunt (right to left shunt)
- Results in **late cyanosis** due to passing deoxygenated blood to the **systemic circulation**
- Could develop in 1-2 years in large defects, or as long as a few decades in small defect

Eisenmenger's Syndrome



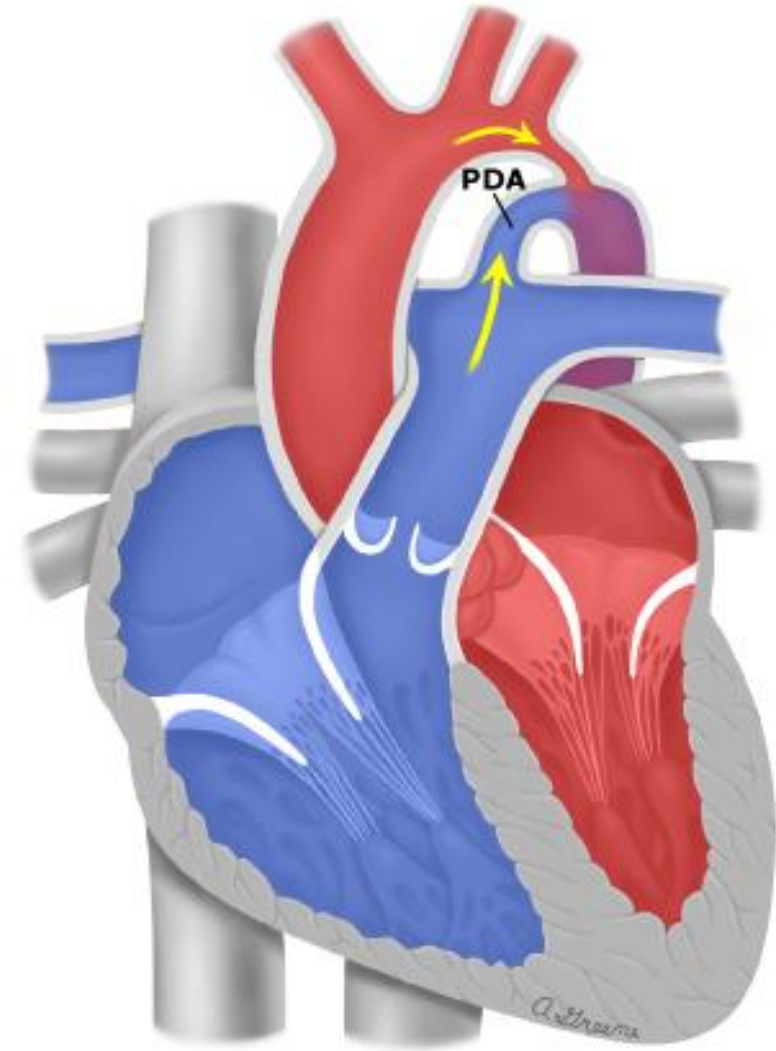
Ectopia Cordis

- Heart is in an abnormal location
- Thoracic ectopia cordis →
 - Heart is partly or completely **exposed on the thoracic wall.**
 - Faulty development of the sternum and pericardium because of **failure of complete fusion of the lateral folds** in the formation of the thoracic wall during the fourth week
- Abdominal ectopia cordis → Heart protrudes through the diaphragm into the abdomen.
- Death occurs in most cases during the first few days after birth, usually from infection, cardiac failure, or hypoxemia.



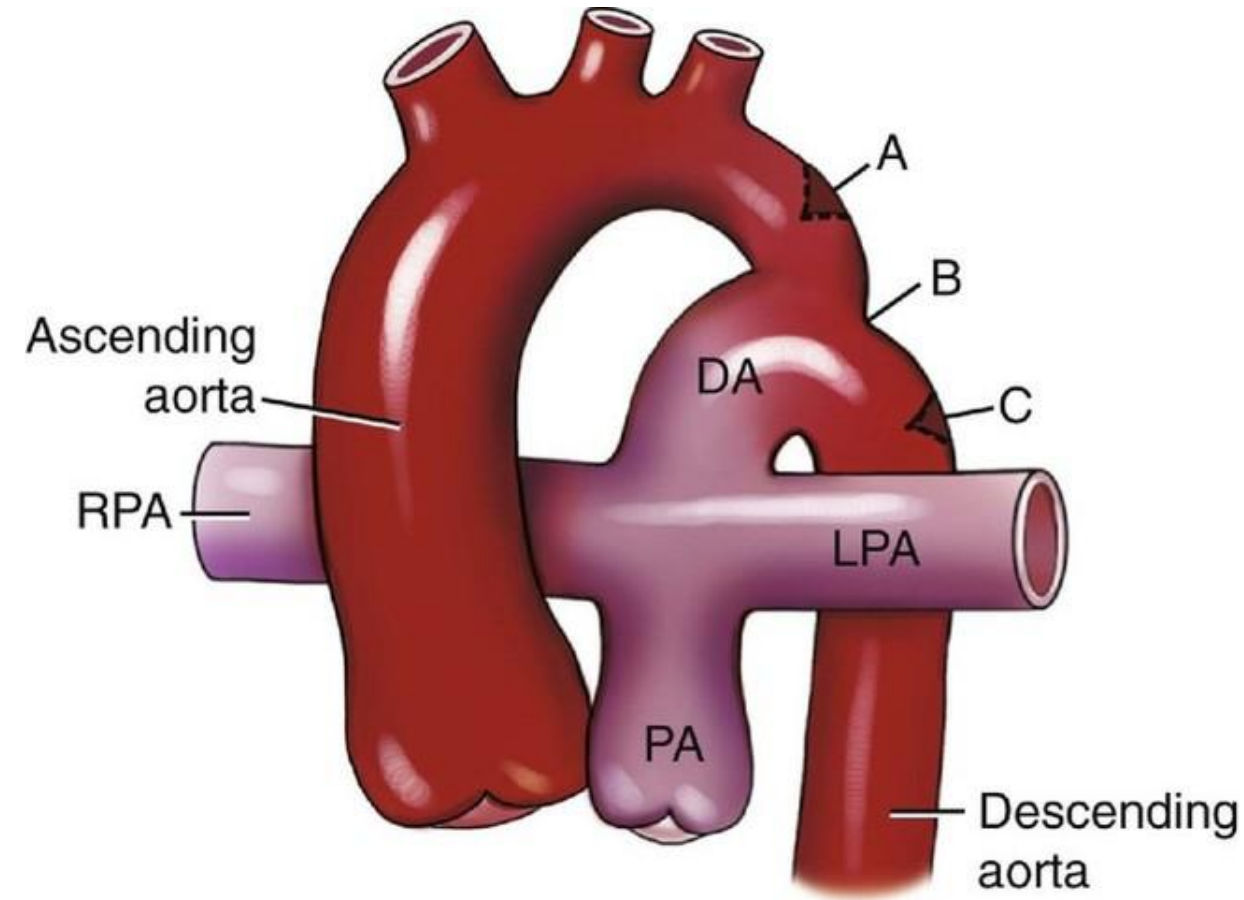
Coarctation of the Aorta

- Congenital narrowing of a segment of the aorta most frequent at the arch of the aorta.
- Male > female (2:1)
- 15-20 % of **Turner syndrome** patients
- Embryological theories:
 - Abnormal involution of a small distal segment of the left dorsal aorta
 - Migration or extension of ductal tissue into the wall of the fetal thoracic aorta



Coarctation of the Aorta

- Can be located at different sites:
- Preductal** - Proximal to ductus arteriosus
 - Juxtaductal** - Directly opposite the ductus arteriosus
 - Postductal** - Distal to ductus arteriosus
- CoA does not cause a hemodynamic problem in utero, because cardiac output → PDA → descending aorta, (bypassing the site of constriction).
 - During the neonatal period, when the PDA and foramen ovale begin to close → increase in the cardiac output that must cross the narrowed aortic segment to reach the lower extremities → pressure gradient

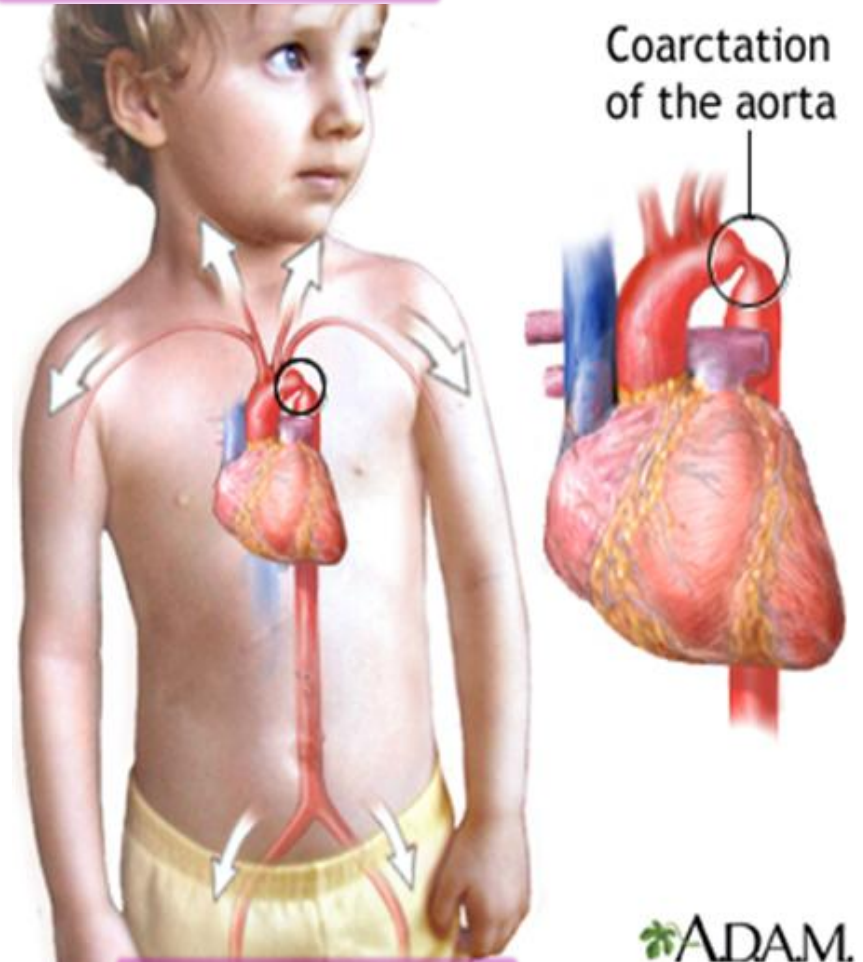


Coarctation of the Aorta

- **Clinical Presentation**

- Patient may complain of **headache, fatigue**
- Underdeveloped lower extremities, which may be cyanotic
- Disparity between **pulses** and **blood pressure** in **upper and lower limbs**
 - Extremely strong pulse and hypertension in the upper limb
 - Delay between **femoral and brachial/radial pulse**
- Imaging studies: **rib notching**, (enlarged collaterals), cardiomegaly.
- Management:
 - Prostaglandins in neonates to keep the ductus arteriosus open.
 - Surgical intervention to correct defect.

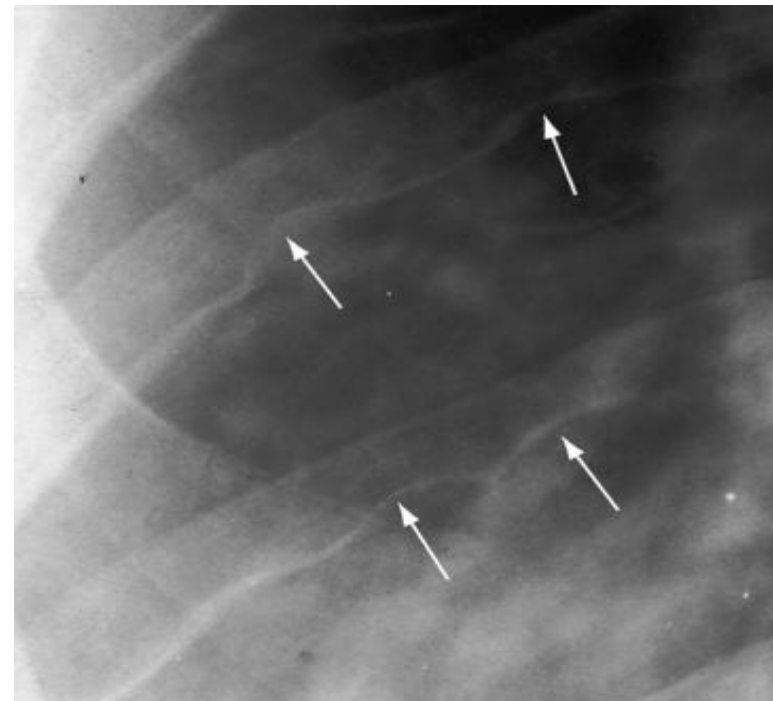
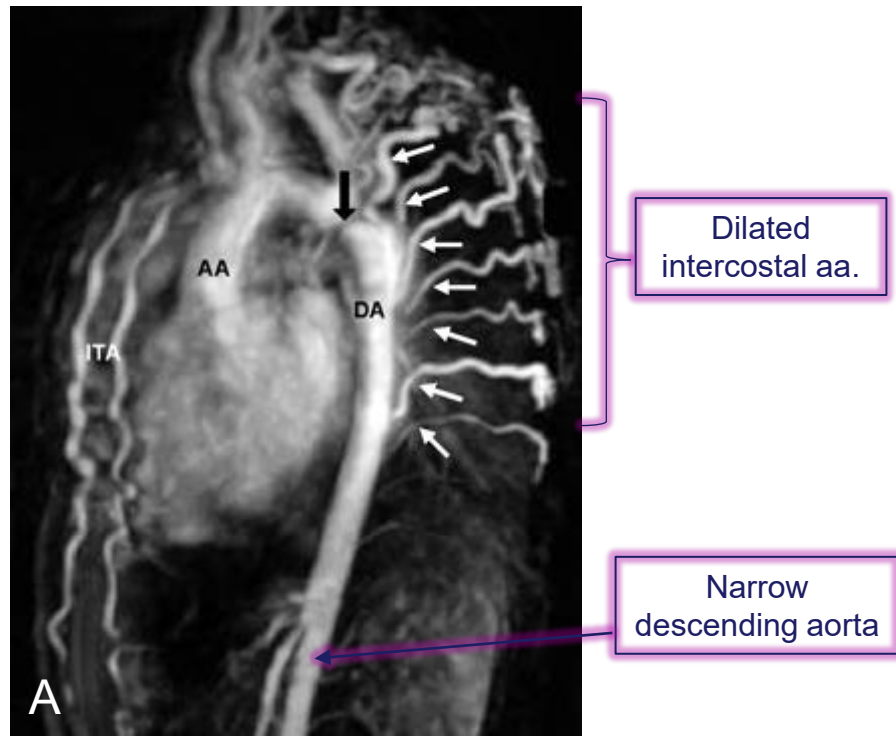
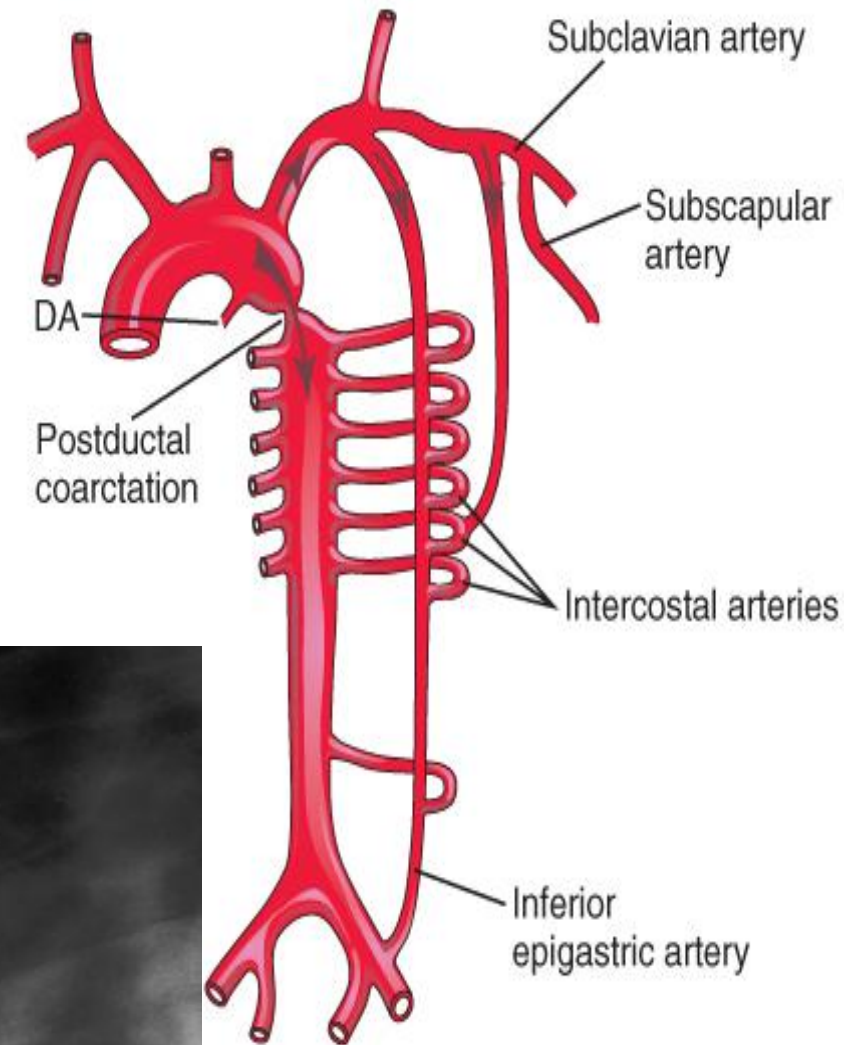
High blood pressures
before the coarctation



low blood pressures
after the coarctation

Coarctation of the Aorta (CoA)

- Collateral Pathway:
- Anterior intercostal arteries (branches of internal thoracic artery) will shunt blood through the subclavian system of arteries as follows:
- **Subclavian → Internal thoracic → Anterior intercostals → Posterior intercostals → Descending aorta**



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